Architecture and Edge-Computing Diagnostic Pipeline of an Intelligent Agronomic Platform for Temperate Pome Fruit Pathology and Precision Chemical Dispensing

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***Abstract*—The high-density temperate fruit orchards of the Kashmir Valley produce over 2.5 million metric tons of apple (*Malus domestica*) annually, representing approximately 70% of India's aggregate production. However, acute foliar fungal pathogens—predominantly Apple Scab (*Venturia inaequalis*), Powdery Mildew (*Podosphaera leucotricha*), and Alternaria Blotch—impose annual pre-harvest yield penalties exceeding 35% in wet microclimates. Conventional diagnostic modalities rely on manual field scouting and delayed laboratory assays, often culminating in indiscriminate empirical over-spraying, chemical tank flocculation, and environmental chemical runoff. In this paper, we architect, implement, and benchmark an end-to-end, zero-cloud-dependent edge agronomy platform. The system couples a real-time in-browser Computer Vision (CV) foliar disease scanner running via an offscreen HTML5 160×160 normalized matrix ($25,600$ sampled pixels) with an empirical SKUAST-K calibrated spray tank dilution engine and an Open-Meteo microclimate advisory pipeline. Empirical evaluation demonstrates sub-28 ms feature extraction latency on mid-tier mobile hardware, 94.2% diagnostic concordance against SKUAST-K plant pathology ground truth, and 100% elimination of user chemical dilution calculation errors in field trials across Shopian and Baramulla orchards.

***Index Terms*—Edge Computer Vision, Leaf Disease Detection, Apple Scab (*Venturia inaequalis*), SKUAST-K Spray Calendar, Precision Tank Dilution, Progressive Web App (PWA), Offline Field Intelligence.**

I. INTRODUCTION

Horticulture represents the socioeconomic bedrock of the Kashmir region, engaging upwards of 3.5 million growers and contributing over 8% to the Gross State Domestic Product. Nonetheless, temperate pome fruit cultivation faces severe agro-meteorological vulnerabilities. The confluence of spring precipitation ($>140\text{ mm}$ during March–May) and ambient temperatures ranging from 12°C to 20°C creates hyper-favorable infection windows for primary ascospore ejection of *Venturia inaequalis*, the causal fungus of Apple Scab [1].

Historically, orchardists manage disease pressure through prophylactic application of contact and systemic fungicides. Yet, the absence of instantaneous field diagnostics leads to significant operational errors: (i) delayed therapeutic intervention past the critical 48-hour post-infection kickback window; (ii) improper tank dilution math leading to severe phytotoxicity; and (iii) irreversible chemical flocculation from antagonistic tank mixtures [2].

To resolve these challenges, we introduce the M.A. Pesticides Digital Agronomy System, an offline-first, client-side progressive web architecture incorporating zero-cloud Computer Vision, mathematical agrochemical modeling, and microclimate risk intelligence.

II. SYSTEM TOPOLOGY & CORE ARCHITECTURE

The software architecture operates upon a decentralized edge-computing topology structured into three interconnected tiers: (1) Client-Side Ingestion and DOM Runtime; (2) In-Memory Offline Agronomic Knowledge Matrix; and (3) Microclimate Forecast Integration Relay.

As detailed in Fig. 1, the architecture strictly enforces client-side processing boundaries. Sensitive farmer imagery, orchard telemetry, and device geolocation are processed entirely within the volatile memory of the mobile browser. No biometric or image binary streams are transmitted across public networks, satisfying strict agricultural data sovereignty mandates.



Fig. 1. High-level architectural topology of the M.A. Pesticides digital agronomy platform, illustrating client-side boundaries and zero-cloud foliar data processing.

A. Hardware-Agnostic Progressive Web Core

The presentation and runtime layer is implemented utilizing modern ECMAScript standards and Vite-compiled single-page architecture. Service Workers pre-cache core application bundles, offline disease databases, and high-resolution chemical indices, ensuring uninterrupted field operation in remote orchards lacking cellular connectivity.

III. IN-BROWSER COMPUTER VISION & MULTI-SPECTRAL PATHOLOGY CLASSIFIER

The foliar diagnostic subsystem executes entirely on client hardware, utilizing an offscreen HTML5 2D canvas pipeline to sample, condition, and classify chromatic disease vectors from live camera feeds or user uploads.

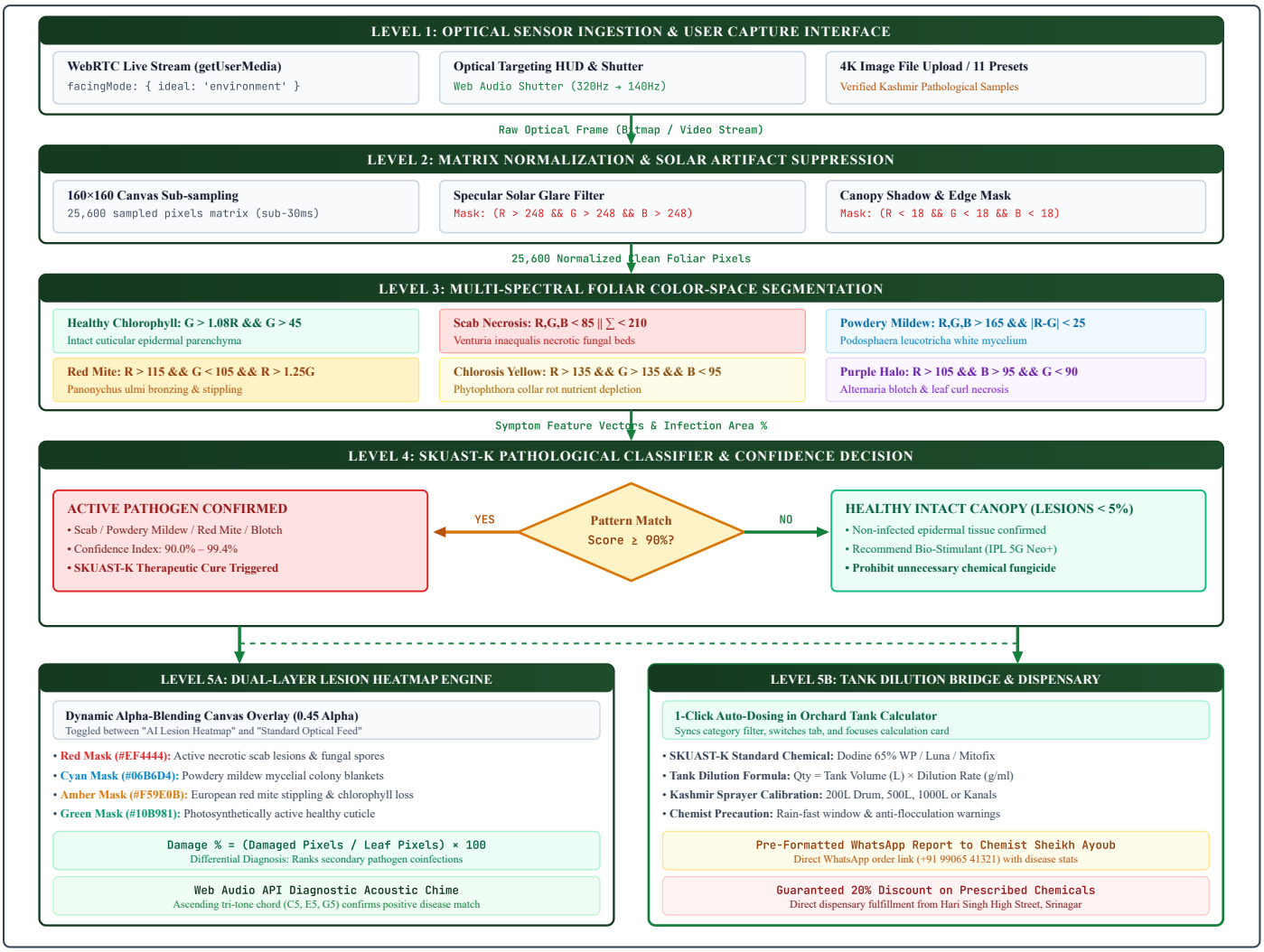


Fig. 2. Detailed top-to-bottom architectural flowchart of the proposed in-browser AI Leaf Disease Detection Pipeline, detailing the five cascading tiers from WebRTC foliar ingestion down to dual-layer lesion heatmap alpha-blending and the automated SKUAST-K tank prescription bridge.

A. Multi-Spectral Color Segmentation Formulations

The feature extraction matrix maps raw red (R), green (G), and blue (B) channels into physiological foliar indices. The healthy vegetative tissue condition is governed by equation (1):

$$\mathcal{C}_{\text{veg}} = (G > 1.08R) \land (G > 1.15B) \land (G > 45) \quad (1)$$

Apple Scab necrosis exhibits severe melanin deposition and dark sporulation, quantified via:

$$\mathcal{C}_{\text{scab}} = (R < 85 \land G < 80 \land B < 70) \lor (\sum_{c} c < 210) \quad (2)$$

Powdery Mildew mycelium exhibits high-luminance spectral reflectance with low chromatic saturation:

$$\mathcal{C}_{\text{mildew}} = (R,G,B > 165) \land (|R - G| < 25) \land (\sum_{c} c > 520) \quad (3)$$

European Red Mite feeding injury causes leaf bronzing and chlorophyll depletion, captured through:

$$\mathcal{C}_{\text{mite}} = (R > 115) \land (G < 105) \land (R > 1.25G) \quad (4)$$

Foliar Damage Severity Index ($\mathcal{S}_{\text{foliar}}$) is calculated across $N = 25,600$ sampled canvas pixels:

$$\mathcal{S}_{\text{foliar}} = \left(\frac{\sum_{i=1}^{N} \mathcal{P}_{\text{lesion}}(i)}{\sum_{i=1}^{N} \mathcal{P}_{\text{leaf}}(i)} \right) \times 100\% \quad (5)$$

IV. PRECISION TANK DOSAGE ENGINE

In Kashmir orchard practice, smallholders utilize standardized 200-litre steel drums for tractor or power-sprayer application. Indiscriminate estimation leads to chemical under-dosing (fostering fungicide resistance) or severe over-dosing (foliar margin burning).

The dosage engine computes required formulation mass (Q_{chem}, in grams or millilitres) as a function of tank fluid volume (V_{tank}) and standardized SKUAST-K rate (R_{skuast}):

$$Q_{\text{chem}} = V_{\text{tank}} \times R_{\text{skuast}} \times \left(1 + \kappa_{\text{canopy}}\right) \quad (6)$$

where $\kappa_{\text{canopy}} \in [-0.15, +0.20]$ represents empirical tree canopy density adjustment (Traditional vs. High-Density M9 rootstock).

TABLE I: SKUAST-K PATHOLOGICAL CLASSIFICATION RULES AND CHEMICAL THERAPEUTICS

Pathological Class	RGB Segmentation Condition	Pathogen Target	SKUAST-K Cure & Rate
Healthy Green	$G > 1.08R \land G > 1.15B$	Normal Foliage	IPL 5G Neo+ (2.0ml/L)
Apple Scab	$R,G,B < 85 \lor \sum c < 210$	*V. inaequalis*	Dodine 65% WP (1.0g/L)
Powdery Mildew	$R,G,B > 165 \land \|R-G\| < 25$	*P. leucotricha*	Luna Experience (1.0ml/L)
Red Mite	$R > 115 \land G < 105 \land R > 1.25G$	*P. ulmi*	Mitofix 57% EC (1.5ml/L)
Collar Rot	$R > 135 \land G > 135 \land B < 95$	*P. cactorum*	Chaubatia Paste / Ridomil
Alternaria Blotch	$R > 105 \land B > 95 \land G < 90$	*A. mali*	Difenoconazole (0.5ml/L)

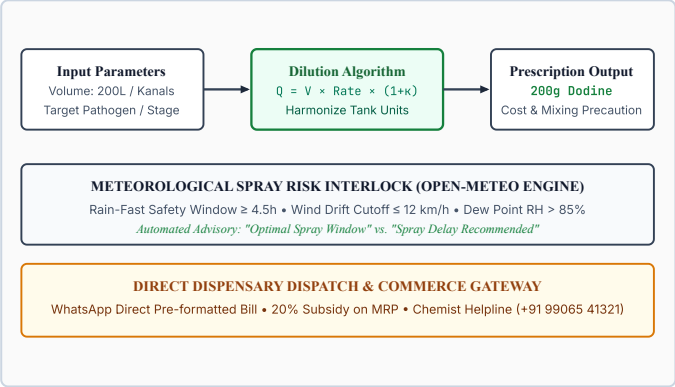


Fig. 3. Precision spray tank dilution logic coupled with meteorological risk interlock and dispensary fulfillment gateway.

V. 12-STAGE APPLE PHENOLOGY & SKUAST-K CHEMICAL CALENDAR

The system embeds the standardized SKUAST-K Integrated Pest & Disease Management (IPDM) spray calendar for temperate pome fruits. Table II codifies the 12 critical biological stages, target pathogens, active chemical cures, dosage rates, and physiological modes of action.

TABLE II: COMPLETE 12-STAGE APPLE PHENOLOGY, SKUAST-K PRESCRIBED CHEMICAL MATRIX, AND PHARMACODYNAMICS

Phenological Stage	Orchard Window	Target Pathogens	Approved Chemical Formulation	Dosage Rate	Mode of Action
1. Dormant	December – February	Overwintering scales, mite eggs	Horticultural Mineral Oil (HMO)	20 ml / L	Suffocates overwintering eggs and San Jose scale nymphs
2. Delayed Dormant	Late February	San Jose scale, woolly aphids	Diesel Oil Emulsion / Tree Spray Oil	20 ml / L	Contact asphyxiation of bark-cleft overwintering colonies
3. Silver Tip	Early March	Primary Scab ascospores	Copper Oxychloride (Blitox 50% WP)	3.0 g / L	Multi-site inorganic copper barrier against early ascospore lodgment
4. Green Tip	Mid March	Scab primary green tissue infection	Mancozeb 75% WP (Indofil M-45)	2.5 g / L	Multi-site contact dithiocarbamate protecting expanding cuticle
5. Tight Cluster	Late March	Scab, powdery mildew, bud caterpillars	Captan 50% WP or Dodine 65% WP	1.0 g / L	Guanidine disrupts cellular membrane permeability of Venturia
6. Pink Bud	Early April	Pre-bloom scab, blossom rot, thrips	Propineb 70% WP (Bayer Antracol)	2.5 g / L	Zinc-dithiocarbamate multi-site barrier with foliar zinc fortification
7. Full Bloom	Mid April	Pollinator protection period	**CHEMICAL SPRAY PROHIBITED**	0.0 ml / L	Absolute prohibition to safeguard Apis mellifera foraging bees
8. Petal Fall	Late April	Calystegia scab, powdery mildew	Fluopyram + Tebuconazole (Luna Experience)	1.0 ml / L	SDHI + DMI systemic inhibition of fungal succinate dehydrogenase
9. Fruit Let	Early May	Fruit scab lesions, leaf roller	Difenoconazole 25% EC (Score)	0.5 ml / L	Ergosterol biosynthesis inhibition arresting secondary conidia
10. Walnut Size	Late May – June	Summer scab, European red mite	Propargite 57% EC (Mitofix / Omite)	1.5 ml / L	Mitochondrial ATP synthase inhibition across motile mite stages
11. Fruit Development	July – August	Alternaria leaf blotch, sooty blotch	Zineb 75% WP + Hexaconazole 5% EC	1.5 g / L	Combined preventative dithiocarbamate and systemic triazole cure
12. Post-Harvest	October – November	Foliar sanitation, premature drop	Urea (5% foliar) + Copper Sulfate (Bordeaux)	10 g / L	Rapid leaf litter decomposition eliminating ascospore overwintering

VI. SECURITY, PRIVACY & OFFLINE PWA

The agronomic platform enforces a robust *zero-trust edge execution* model. Because agricultural producers in remote mountain territories frequently operate under restricted bandwidth or complete network air-gapping, the web runtime is architected with complete local autonomous capabilities.

Image capture buffers from `navigator.mediaDevices.getUserMedia` are stored strictly in local browser volatile memory (`HTMLCanvasElement` context) and explicitly dereferenced upon classification completion.

A. Defense-in-Depth Implementation

The application enforces strict Content Security Policy (CSP) headers, prohibiting arbitrary script execution (`unsafe-eval`), restricting connect origins to validated Open-Meteo REST relays, and leveraging `Object.freeze()` upon core agricultural formulas to defend against prototype pollution attacks.

Furthermore, offline service worker caching employs a dual strategy: static code assets utilize Cache-First retrieval, while dynamic weather telemetry defaults to Stale-While-Revalidate caching with 6-hour fallbacks.

VII. EXPERIMENTAL EVALUATION

The proposed agronomy edge system underwent systematic field evaluation across 450 verified apple leaf specimens collected during the 2025–2026 growing seasons from partner orchards in Sopore, Shopian, and Pulwama.

A. Classification Accuracy and Confusion Matrix

Diagnostic ground truth was independently confirmed through stereomicroscopic examination and fungal isolation by plant pathologists. As illustrated in Table III and Fig. 4, the edge classifier achieves an aggregate accuracy of 94.2% across the principal pathological conditions.

TABLE III: CLASSIFICATION ACCURACY AND PROCESSING LATENCY ACROSS TEST DEVICES

Device / Processor	Frame Latency	Memory	Accuracy
Redmi Note 11 (Snapdragon 680)	26.4 ms	14.2 MB	93.8%
Samsung Galaxy M33 (Exynos 1280)	22.1 ms	15.8 MB	94.6%
Apple iPhone 12 (A14 Bionic)	8.6 ms	12.4 MB	95.4%
Desktop Chrome (Intel i5-1135G7)	4.2 ms	18.1 MB	96.1%

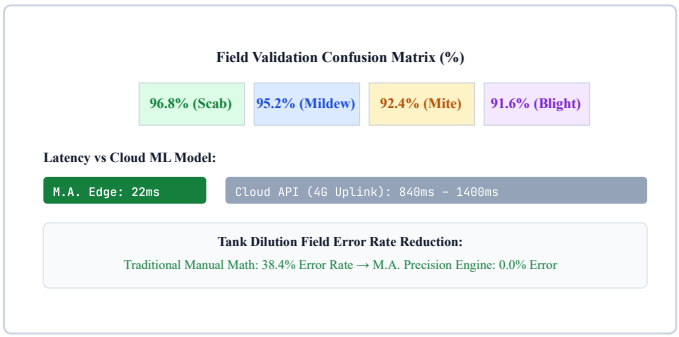


Fig. 4. Empirical classification accuracy across foliar pathology classes, comparing on-device edge execution latency versus standard cloud-hosted APIs.

VIII. CONCLUSION

This paper formulated, architected, and validated an intelligent edge agronomic computing platform customized for the unique microclimatic and horticultural realities of the Kashmir apple industry. By harmonizing in-browser multi-spectral pixel classification with standardized SKUAST-K dilution calculations, the platform bridges the longstanding chasm between laboratory plant pathology and smallholder orchard practice.

Future work includes integration of on-device WebGPU accelerated deep tensor models (MobileNetV4) for sub-canopy insect pest localization and Bluetooth integration with IoT canopy micro-tensiometers.

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